

The second-order Taylor bound for the kernel amplitude

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Abstract

For the amplitude $F(u, s) = \eta(u)e^{\beta s \xi(u)}$ with ξ, η admitting first-order Taylor expansions at 0 with quadratic remainders on $[0, b]$, we prove $|F(u, s) - F(0, s) - u \partial_u F(0, s)| \leq C u^2 (1+s)^2 e^{3\beta L s}$ for $u \in [0, b]$, $s \geq 0$, with $\partial_u F(0, s) = e^{\beta s a}(\eta_1 + \beta s \eta_0 a_1)$ and an explicit C . This is the input for the second-order asymptotics of the chart integral (Astra's step 3). Zero `sorry/axiom`.

1 The things formalised

```
theorem abs_exp_sub_one_sub_id_le' (t : ℝ) :
  |Real.exp t - 1 - t| ≤ 3 * t ^ 2 * Real.exp |t|
theorem abs_exp_sub_one_le' (t : ℝ) : |Real.exp t - 1| ≤ |t| * Real.exp |t|
theorem kernel_taylor_bound (b L K M M L a a : ℝ) ( : ℝ → ℝ) (h : 0 < )
  (hb : 0 < b) (hK : 0 < K) (hM : 0 < M) (hM : 0 < M) (hL : 0 < L)
  (hL : u Icc (0 : ) b, | u | ≤ L) (h : | | ≤ M) (h : | | ≤ M) (ha : |a| ≤ L)
  (hT : u Icc (0 : ) b, | u - a - u * a | ≤ K * u ^ 2)
  (hT : u Icc (0 : ) b, | u - - u * | ≤ K * u ^ 2)
  (u s : ℝ) (hu : u Icc (0 : ) b) (hs : 0 ≤ s) :
  | u * Real.exp ( * s * u ) - * Real.exp ( * s * a )
    - u * (Real.exp ( * s * a ) * ( + * s * * a )) |
  (K + M * (L + K * b) * + 3 * M * ^ 2 * (L + K * b) ^ 2 + * K * M)
  * u ^ 2 * (1 + s) ^ 2 * Real.exp (3 * * L * s)
```

2 Proof strategy

Split $F(u, s) - F(0, s) - u \partial_u F(0, s) = (\eta - \eta_0 - u \eta_1) e^{\beta s \xi} + u \eta_1 (e^{\beta s \xi} - e^{\beta s a}) + \eta_0 (e^{\beta s \xi} - e^{\beta s a} - \beta s u a_1 e^{\beta s a})$ and write $e^{\beta s \xi(u)} = e^{\beta s a} e^t$ with $t = \beta s (\xi(u) - a)$, $|t| \leq \beta s X u$, $X = L_1 + K b$. The third piece is $e^{\beta s a} (e^t - 1 - t) + e^{\beta s a} \beta s (\xi - a - u a_1)$; the elementary bounds $|e^t - 1| \leq |t| e^{|t|}$ (from the unit-76 mean-value bound) and $|e^t - 1 - t| \leq 3t^2 e^{|t|}$ (Mathlib's bound for $|t| \leq 1$, trivial otherwise) together with $e^{\beta s a} e^{|t|} \leq e^{3\beta L s}$ give the result; finally $1, s, s^2 \leq (1+s)^2$.

3 Roadblocks and resolutions

- `gcongr` on $3t^2 \cdot (E_0 e^{|t|}) \leq 3(\beta s X u)^2 \cdot E_x$ tries to prove $0 \leq t$; use `sq_abs` and `pow_le_pow_left` on $|t|$ explicitly.
- In a large context `nlinarith` for $s^2 \leq (1+s)^2$ hits the heartbeat limit; prove such facts by the dedicated lemmas (`one_le_pow`, `le_self_pow`, `pow_le_pow_left`) and feed `linarith` the four product inequalities explicitly.

4 Outcome

The kernel Taylor bound is ready to be fed into the density-transfer theorem. Next: the one-dimensional second-order theorem $Z(n) = C_0 n^{-p/2} + C_1 n^{-q/2} + O(n^{-(h+3)/(2k)})$ with $C_1 = \frac{1}{k}[\eta'(0)S_q(a) + \beta\eta(0)\xi'(0)S_{q+1}(a)]$.